
View Displays

About this Chapter

This chapter describes CADD's view-display functions. These functions allow you to display different views of a CADD drawing. They make it possible to work on various portions of a drawing. You can move the image on the screen to display a desired view and enlarge or reduce the images as required.

Key Terms in this Chapter

Term	Description
Pan	To move the view in a specific direction.
Zoom in	To get a closer view.
Zoom out	To get a view from a greater distance.
Zoom factor	To enlarge or reduce the view by a specific degree.
Viewport	A portion of the drawing window that contains an independent view of the drawing.
Window	An imaginary rectangle formed by the indicated points.

The View-Display Functions

CADD's view-display functions enable you to display different views of the drawing. You need to constantly change views in order to work on different portions of the drawing. For example, to draw a line from one corner of a large drawing to another, you may need to change the three views. First, magnify the corner where the line is to be started. Second, reduce the display to see the entire drawing. Third, magnify the corner to indicate the end point of the line.

Working with the view-display functions is similar to using a telescopic camera. You can zoom out to view larger objects, and zoom in to get a closer view of objects. However, CADD's functions are much more precise and flexible. You can magnify the drawings to an exact degree, store specific views in computer memory, and then display them back when required.

The following are the basic techniques to display views:

- Display a full view
- Move the view
- Zoom-in and zoom-out
- Enlarging view by indicating a window
- Saving and displaying views
- Display previous view

You can enter view-display functions in the command window or by using the tool buttons. Fig. 4.1 shows some examples of tool buttons used to enter view display functions. The exact icons used vary from one program to another.

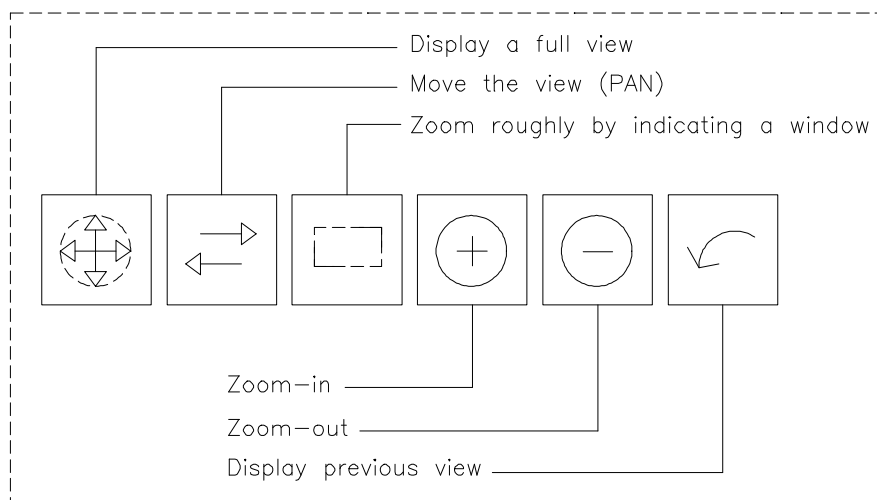


Fig. 4.1: Examples of view-display tool buttons.

To illustrate the view-display functions, we will use an engineering drawing shown in Fig. 4.2. The drawing shows mechanical parts of a machine in minute detail. We will illustrate how to work on this complex drawing using view-display functions.

Note:

The use of view display does not change the size of drawing objects in any way. They are just displayed at a specific magnification level.

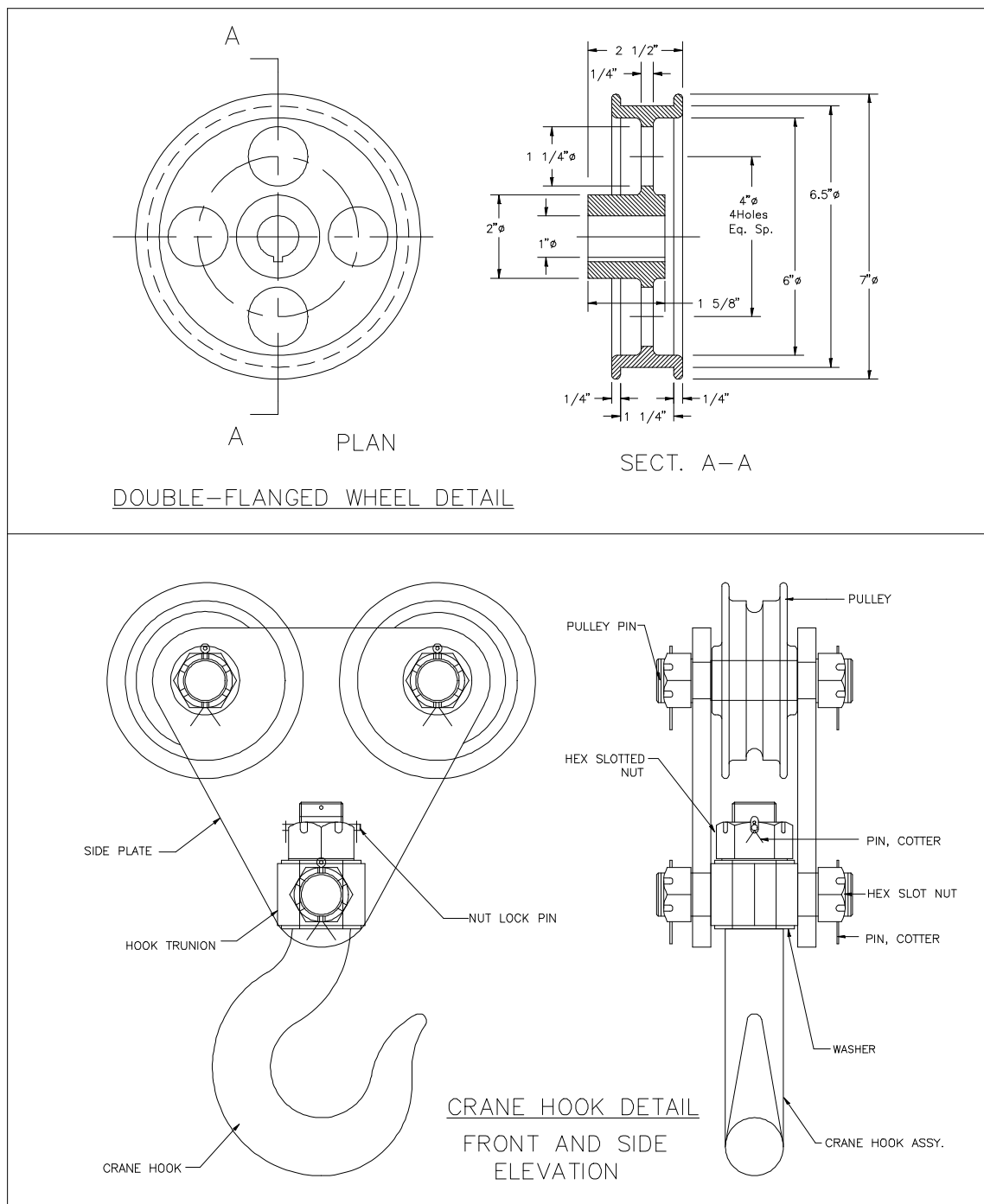


Fig. 4.2: An engineering drawing showing plan, section and elevation of a mechanical part.

Displaying a Full View

CADD allows you to display the entire drawing on-screen regardless the size of the drawing. If the drawing is too large, CADD automatically reduces it to fit the drawing area. If the drawing is too small, CADD automatically enlarges it to fill the drawing area.

Example: If you draw a 1"x1" square and display a full view of the drawing, it magnifies the image to fill the entire screen. Now let's say you draw a 10' diameter circle around the square and then display a full view. It will automatically reduce the entire image to fit that circle on the screen. The square will be reduced significantly and might look like only a dot.

Fig. 4.3 (Screen 1) illustrates how you can display the entire drawing (Fig. 4.2) to fit on-screen by clicking on the full view tool button.

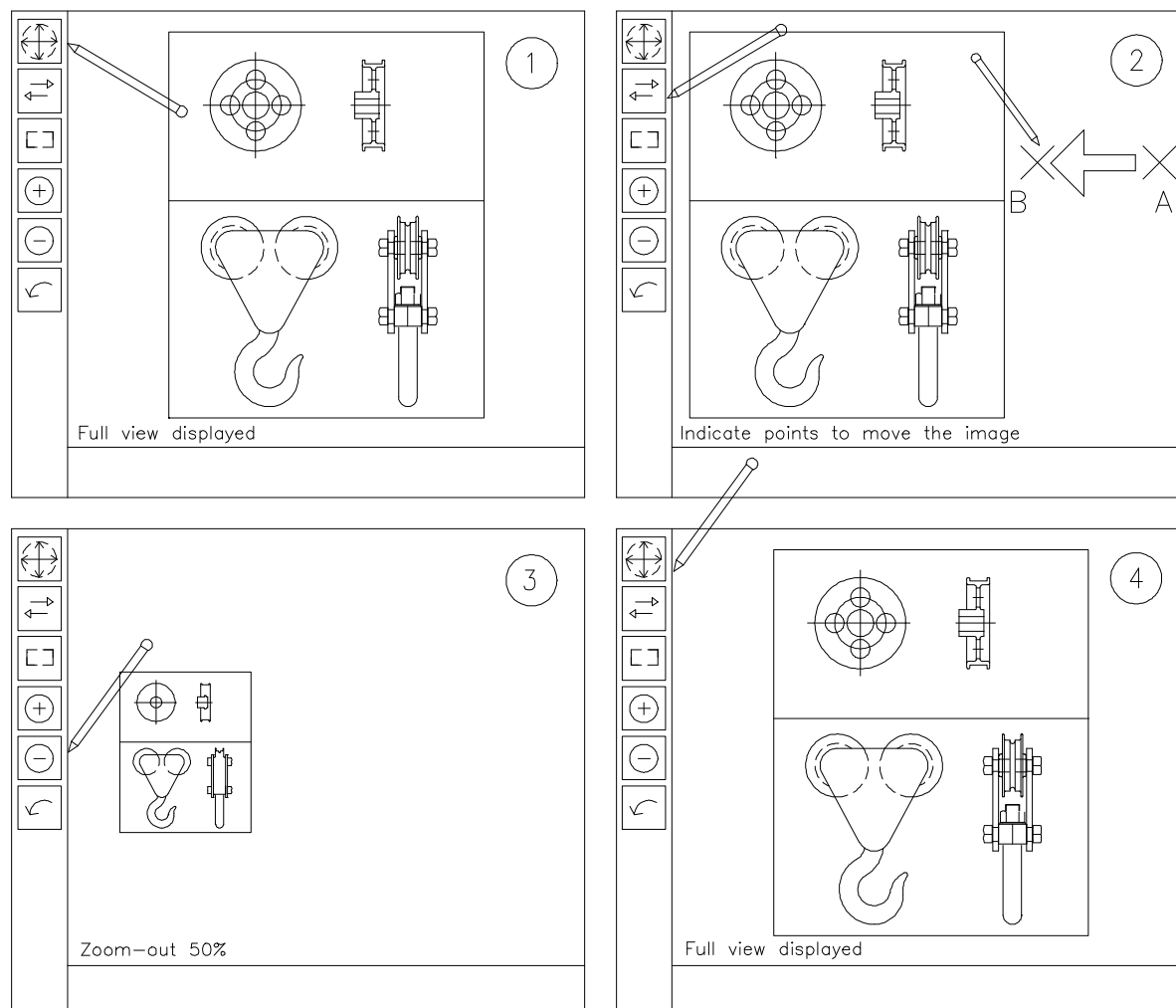


Fig. 4.3: The drawing image can be enlarged, reduced or moved using view-display functions.

Moving the View

You can move the image in any direction with the push of a button. This capability of CADD is often referred to as "pan". To move the image, all you need to do is indicate two points between which the drawing should move. The drawing is moved from the first indicated point to the second indicated point. Screen 2 (Fig. 4.3) shows how you can move the drawing to the left by indicating points A and B.

In some CADD systems the same result is achieved by pressing the arrow keys on the keyboard or by using the scroll bars on the screen. The drawing is moved in the direction of the arrow key pressed. You can preset a percentage of how much the drawing should move with each press of an arrow key.

Zoom-in and Zoom-out

CADD allows you to reduce or enlarge the image view by a specified degree. You can enter a specific percentage or a zoom factor to reduce or enlarge an image. To enlarge the image 1 1/2 times, you need to enter the percentage value as 150% or the zoom factor as 1.5. To reduce the image by half, you need to enter the percentage value as 50% or zoom factor as 0.5.

Most of the CADD programs provide tool buttons that allow you to instantly zoom-in and zoom-out by a predefined percentage. For example, clicking on the zoom-in tool button enlarges the image by 150% and clicking on the zoom-out button reduces the image by 50%.

Screen 3 (Fig. 4.3) shows how you can reduce the image shown on Screen 2 by 50%. Simply click on the zoom-out tool button and the image is reduced as shown.

Enlarging View by Indicating a Window

You often need to enlarge drawing images in order to work with a greater detail. You can specify an area to be enlarged by designating an enlargement window on the screen. A window is an imaginary rectangle formed by two diagonal points. The section of diagram contained within the window is enlarged to fit the entire drawing area.

Fig. 4.4 shows how you can enlarge a portion of the diagram by indicating a window. Screen 1 shows the full view of the drawing. Click on the zoom window tool button to enlarge a corner of the drawing and enter points A and B as the corners of the enlargement window. The view is enlarged as shown on Screen 2. To view the drawing in more detail, enter the zoom window command again and enter points A and B as the corners of enlargement window. The view is enlarged as shown on Screen 3. Similarly, enlarge the view again to see the drawing in more detail as shown on Screen 4.

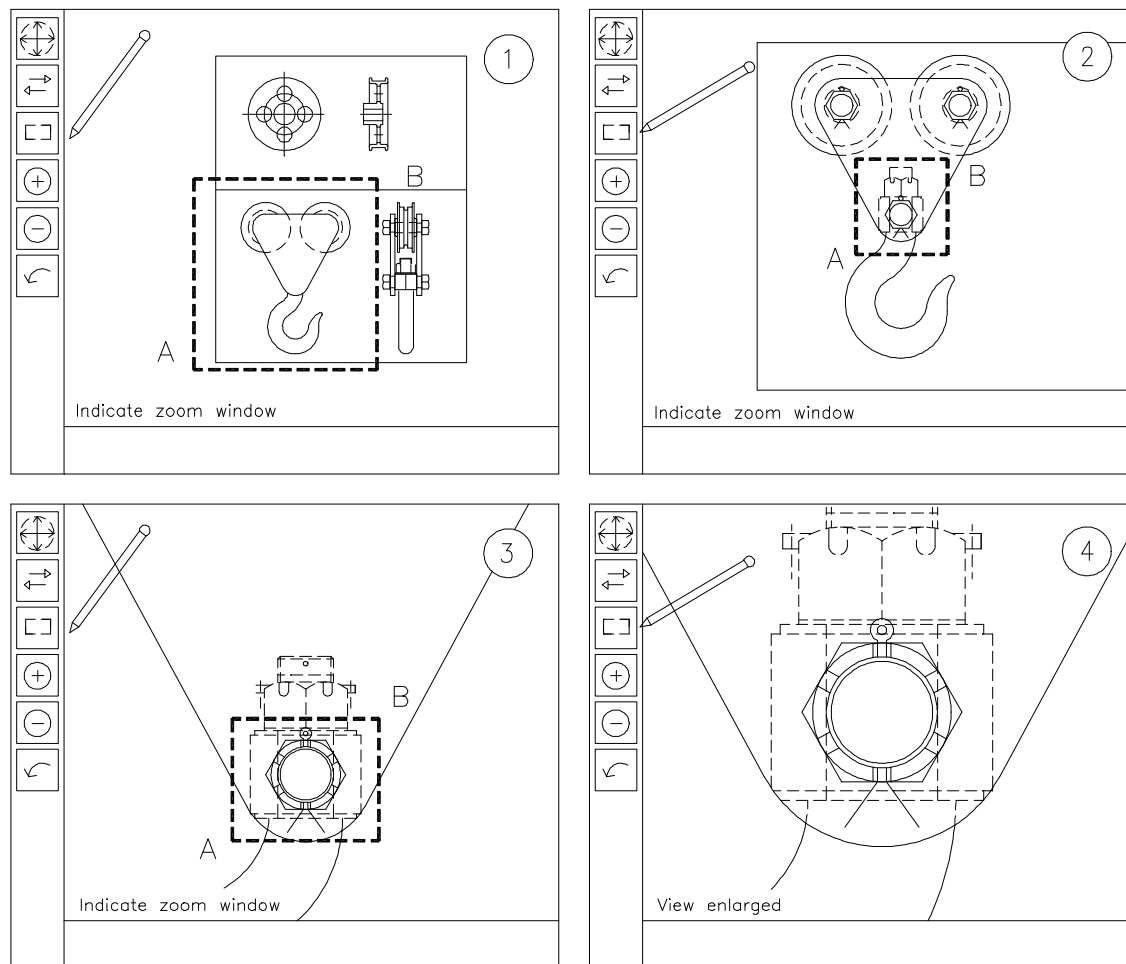


Fig. 4.4: Views of the drawing are enlarged using view-display functions.

Saving and Displaying Views

CADD allows you to store selected views and later display them back when required. When you are working with complex drawings, you need to display a number of views quickly. This function provides a convenient way to display selected views.

When you determine that you need to work on certain portions of a drawing frequently, you can store those views by giving them different names. To display any of the views, just reference the name of the view and it is instantly displayed.

Displaying Previous Views

CADD makes it easy to return to previously displayed views. Just click on the tool button designated to display the previous view and you are returned to the last view displayed. This feature is extremely useful because you often need to return to the portions of the drawing where you were recently working. By using this function, you can display all the previous views one by one by entering the function again and again. Some programs also allow you to display subsequent views after you return to previous views. You can go back and forth between all the views just by clicking on a tool button.

AutoCAD, MicroStation and Cadkey Terms

*The following are the important terms used in leading CADD programs:
(The exact procedures vary from one program to another)*

Task	AutoCAD	MicroStation	Cadkey
Displaying a full view	ZOOM: ALL ZOOM: EXTENTS	FIT VIEW	VIEW: ZOOM AUTOSCALE
Moving the view sideways	PAN	PAN VIEW	VIEW: PAN
Reducing the view	ZOOM (-)	ZOOM OUT	VIEW: MIN
Enlarging the view	ZOOM: WINDOW ZOOM (+)	ZOOM IN	VIEW: MAX
Saving and displaying views	VIEW: SAVE/DISPLAY	VIEW: SAVE/RECALL	DEFINE: DISPLAY VIEW
Displaying the last view	ZOOM: PREVIOUS	VIEW: PREVIOUS	VIEW: BACK 1

Important Tips:

- There are a number of additional view display techniques available in advanced CADD programs. One such technique is called real time pan and zoom that allows you to instantly manipulate the image by moving the cursor in a specific direction.
- Most CADD programs allow you to display multiple views side by side. The drawing area can be divided into different sections called viewports. You can display a different view in each viewport. This is particularly helpful when working with 3D drawings. For example, you can display a plan view in one viewport, elevation in another and a 3D view in another.